

# Lesson 29 — Copy That (improve a custom classifier)

Matches LEGO Teacher Portal lesson: "Copy That".

**Objective (I can...):** I can improve a classifier's accuracy by training it with better examples. **CS concept:** Training quality and accuracy (more/varied examples improve results) · **Builds on:** Lesson 28 (training a classifier). *Privacy: on-device camera, no logins, images never saved.*

**AI Connection (2 min):** This is why we said "more tries help" back in the loops unit — better, more varied examples make AI more accurate. *First-principles question:* "What kind of example would fix a classifier that keeps guessing wrong?"

**Materials:** LEGO CS & AI kit + device per pair with camera. **Model reuse:** pose-reactor model from Lesson 28 — code and retrain only.

**60-minute flow - Warm-up (8):** "Tricky twins" — coach shows two similar gestures; ask how to tell them apart reliably. Connect to clearer, more varied examples. - **Concept + demo (10):** Demo a classifier that guesses wrong, then add more and better example photos; show accuracy improve. - **Build/Code (30):** Pairs train the classifier to reliably tell categories apart — test accuracy, add examples until it's dependable, then link recognitions to model actions. **Keep the data tally going** (examples-per-category) and re-check "are my categories balanced?" *Swap at 15 min.* - **Challenge (7):** Introduce a tricky look-alike category; train until the model handles it. - **Share + discussion (5):** Pairs report how accuracy changed, then discuss **why** the classifier still guesses wrong sometimes: did the training examples "represent everyone/everything"? (E.g., trained in only one kind of lighting, or on only one person, or with lopsided category counts.) Name one better example that would help. Model stays built.

**Pair roles:** Driver captures new examples, Navigator tracks and reports accuracy. Swap at 15 min. (*Trio: third runs the accuracy tests.*)

**Differentiation:** *Support* — focus on improving one category. *Extension / fast-finisher* — balance three categories for high accuracy. *Reteach* — add examples one at a time and re-test so the improvement is visible each time.

**Exit check:** Pair shows measurable improvement after adding/varying training examples, and can name one reason an AI might make a mistake and one way better examples would help. **At-home (optional):** Explain to a family member why AI needs many examples to learn well. ~10 min.